

WebApi Handson Solution

3. WebApi Hands-on Solution

Objectives Addressed

- * Action method to return a list of a custom class entity.
- * Usage of `[FromBody]` attribute.
- * Demonstrate Custom filter (`ActionFilterAttribute`, `ExceptionFilter`).

Solution Implementation

1. Custom Model Classes

```
```csharp
using System;
using System.Collections.Generic;

namespace WebApi_Handson.Models
{
 public class Skill { public string Name { get; set; } }
 public class Department { public string Name { get; set; } }

 public class Employee
 {
 public int Id { get; set; }
 public string Name { get; set; }
 public int Salary { get; set; }
 public bool Permanent { get; set; }
 public Department Department { get; set; }
 public List<Skill> Skills { get; set; }
 public DateTime DateOfBirth { get; set; }
 }
}
```
```

2. Custom Authorization Filter

WebApi Handson Solution

Create a `Filters` folder and add `CustomAuthFilter.cs`.

```
```csharp
using Microsoft.AspNetCore.Mvc;
using Microsoft.AspNetCore.Mvc.Filters;
using System.Linq;

namespace WebApi_Handson.Filters
{
 public class CustomAuthFilter : ActionFilterAttribute
 {
 public override void OnActionExecuting(ActionExecutingContext context)
 {
 if (!context.HttpContext.Request.Headers.TryGetValue("Authorization", out var token))
 {
 context.Result = new BadRequestObjectResult("Invalid request - No Auth token");
 return;
 }

 if (!token.FirstOrDefault().Contains("Bearer"))
 {
 context.Result = new BadRequestObjectResult("Invalid request - Token present but Bearer
unavailable");
 return;
 }

 base.OnActionExecuting(context);
 }
 }
}
```
```

3. Custom Exception Filter

WebApi Handson Solution

```
```csharp
using Microsoft.AspNetCore.Mvc;
using Microsoft.AspNetCore.Mvc.Filters;
using System.IO;

namespace WebApi_Handson.Filters
{
 public class CustomExceptionHandler : IExceptionHandler
 {
 public void OnException(ExceptionContext context)
 {
 // Log exception to file
 File.AppendAllText("error.log", context.Exception.ToString() + "\n");

 // Set exception result
 context.Result = new ObjectResult(new { error = "Internal server error." })
 {
 StatusCode = 500
 };
 context.ExceptionHandled = true;
 }
 }
}
```
```

4. Employee Controller

```
```csharp
using Microsoft.AspNetCore.Mvc;
using System.Collections.Generic;
using WebApi_Handson.Models;
using WebApi_Handson.Filters;
using System;

namespace WebApi_Handson.Controllers
```

## WebApi Handson Solution

```
{
 [Route("api/[controller]")]
 [ApiController]
 [CustomAuthFilter]
 public class EmployeeController : ControllerBase
 {
 private List<Employee> GetStandardEmployeeList()
 {
 return new List<Employee>
 {
 new Employee { Id = 1, Name = "John", Salary = 50000, Permanent = true },
 new Employee { Id = 2, Name = "Smith", Salary = 60000, Permanent = false }
 };
 }

 [HttpGet]
 [ProducesResponseType(200)]
 [ProducesResponseType(500)]
 public ActionResult<IEnumerable<Employee>> Get()
 {
 // Uncomment next line to test CustomExceptionFilter
 // throw new Exception("Testing exception filter!");
 return GetStandardEmployeeList();
 }
 }
}
```